Patient Readmission Analytics

Identifying High-Risk Patients for Proactive Care

Screenshot Evidence Report

SQL | Excel | Power BI | AI-Assisted Analysis

Project Name	30-Day Readmission Risk Dashboard
Course Project	Patient Readmission Analytics: Identifying High-Risk Patients for Proactive Care
Prepared By	Junaid Shaik Mohammed
Scenario Organization	ReadmitGuard Analytics
Tools Used	SQL, Microsoft Excel, Power BI, AI-assisted analysis
Document Type	Professional screenshot evidence and output documentation

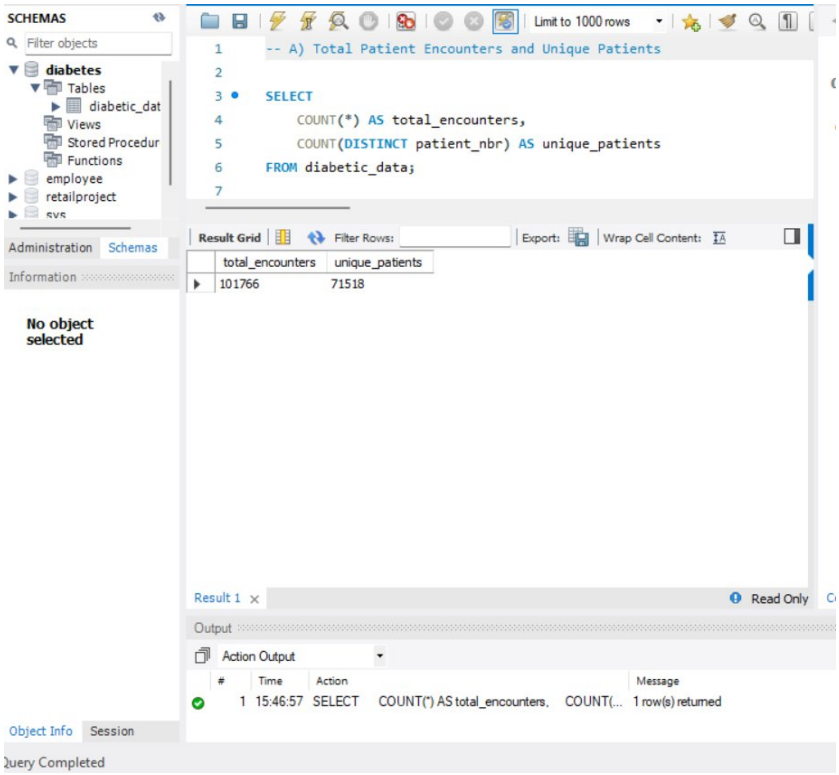
Document Purpose

This report provides structured evidence of the analytical outputs created for the patient readmission capstone project. It documents SQL queries, Excel summaries, pivot charts, AI-assisted interpretation, and Power BI dashboard views used to identify 30-day readmission trends and high-risk patient segments.

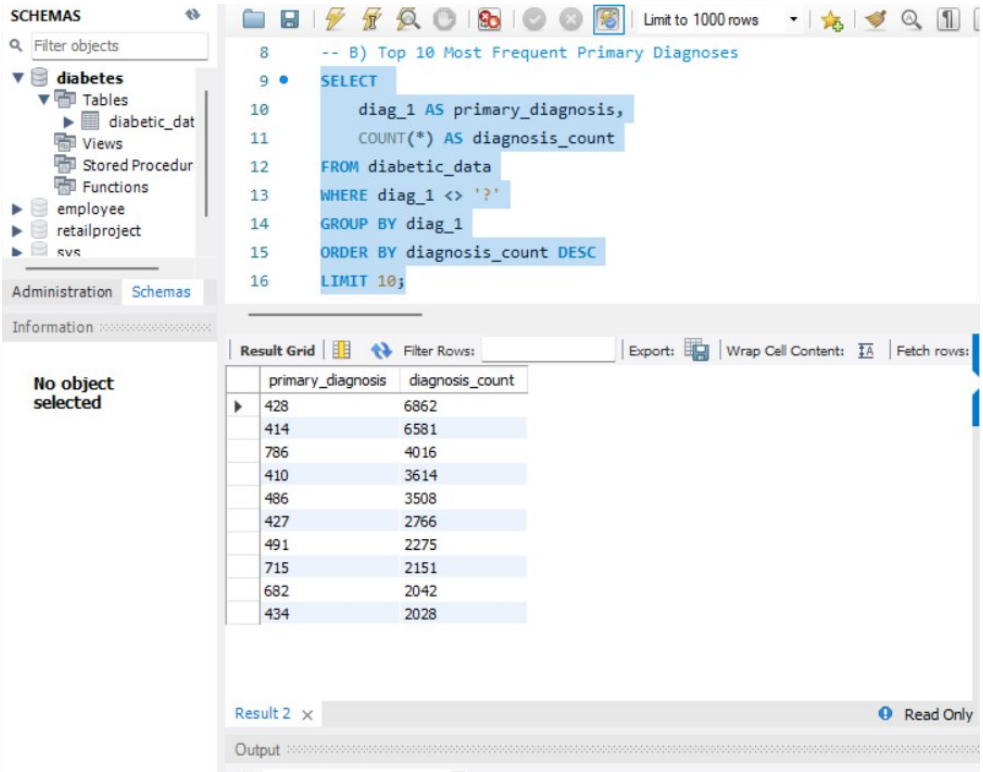
Submission Mapping

Write-up	Business Requirements Document (BRD) submitted separately.
Screenshots	This evidence report is been uploading after PDF conversion.
Source Code / Files	CSV dataset, SQL file, Excel workbook, and Power BI file where available.

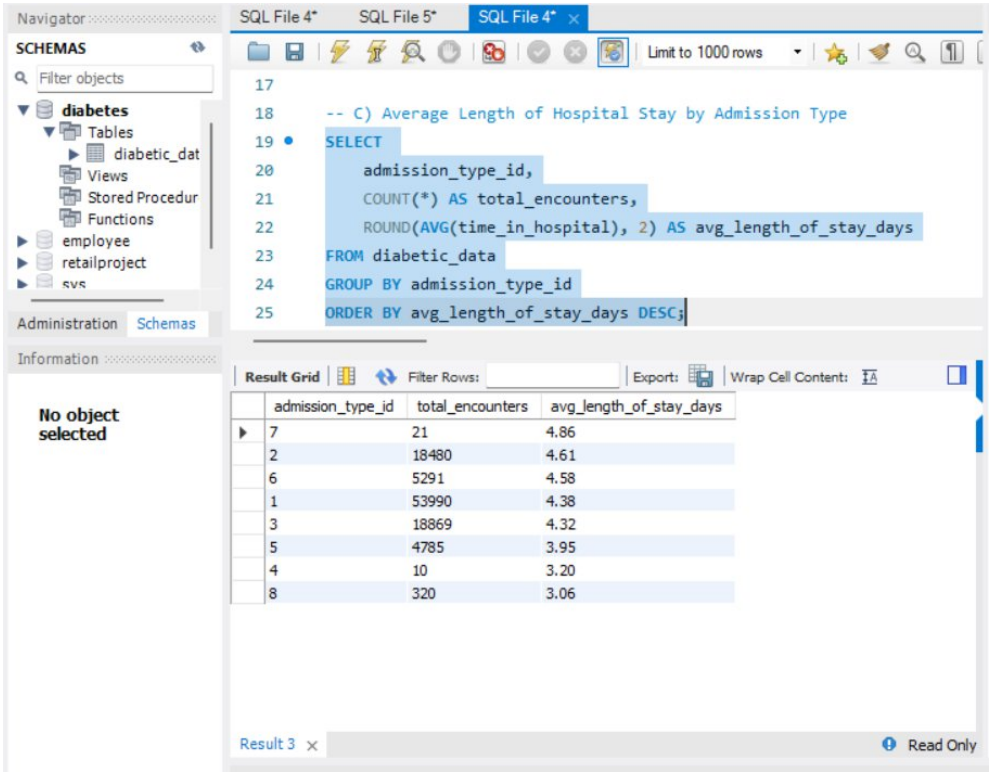
EV-01	SQL – Total Patient Encounters and Unique Patients



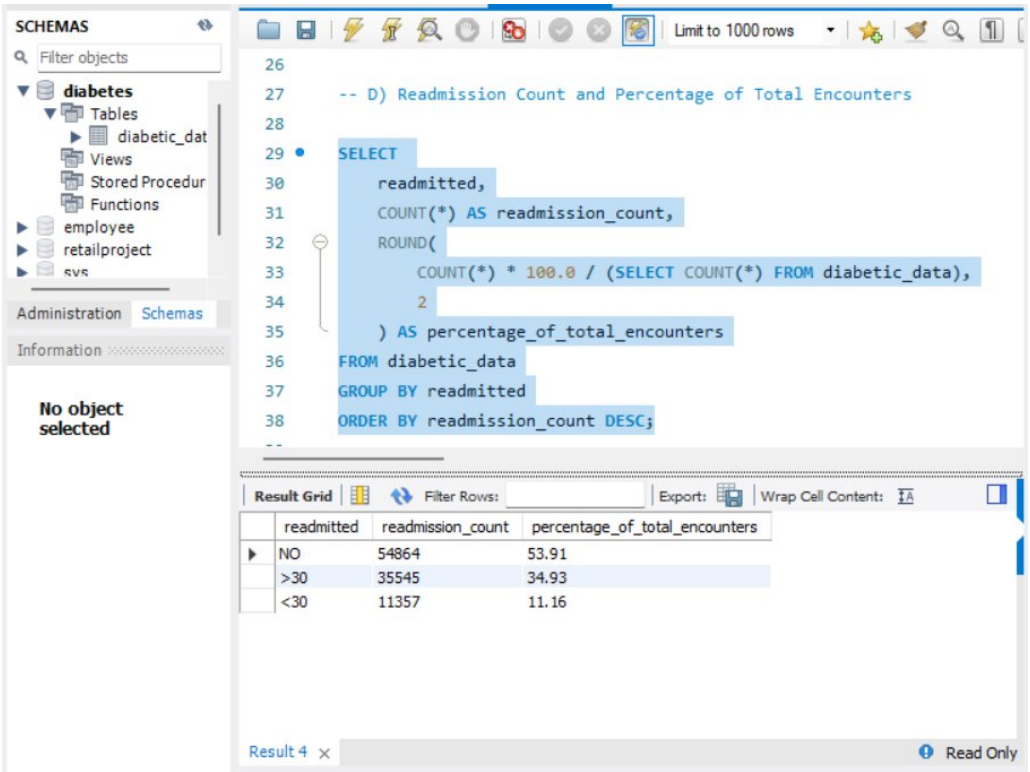
EV-02	SQL - Top 10 Most Frequent Primary Diagnoses



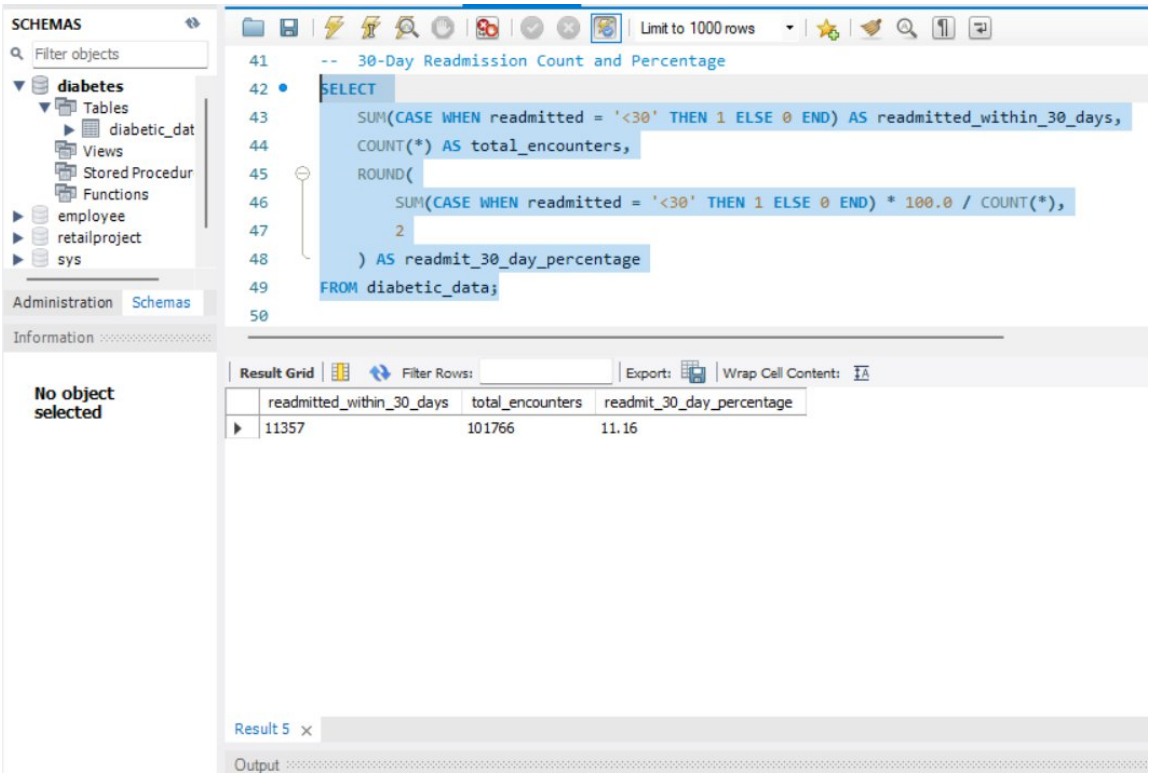
EV-03	SQL – Average LOS in Hospital



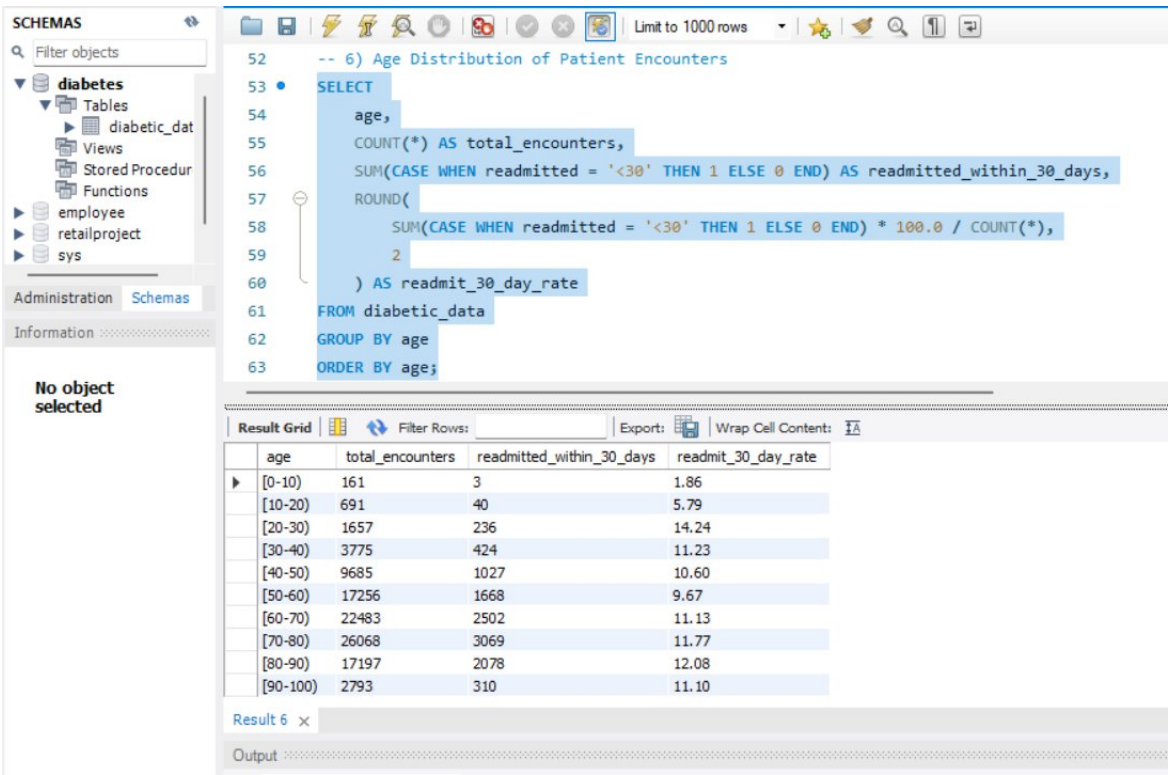
EV-04	SQL – 30-Day Readmission Rate



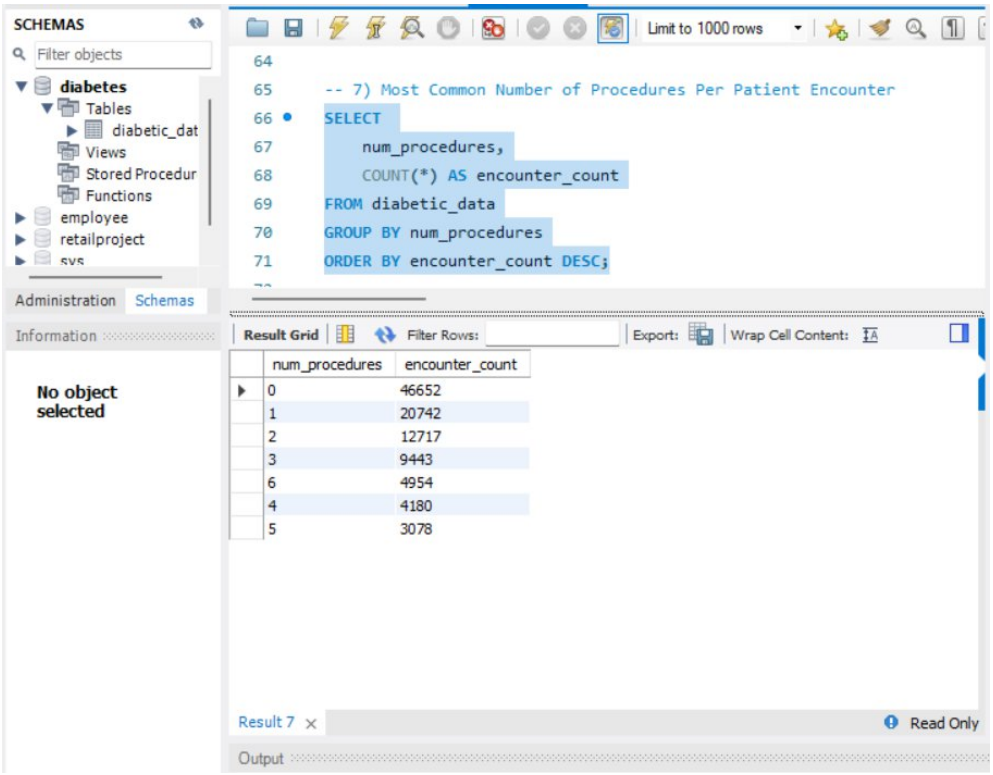
EV-05	SQL – 30-Day Readmission Count & Percentage



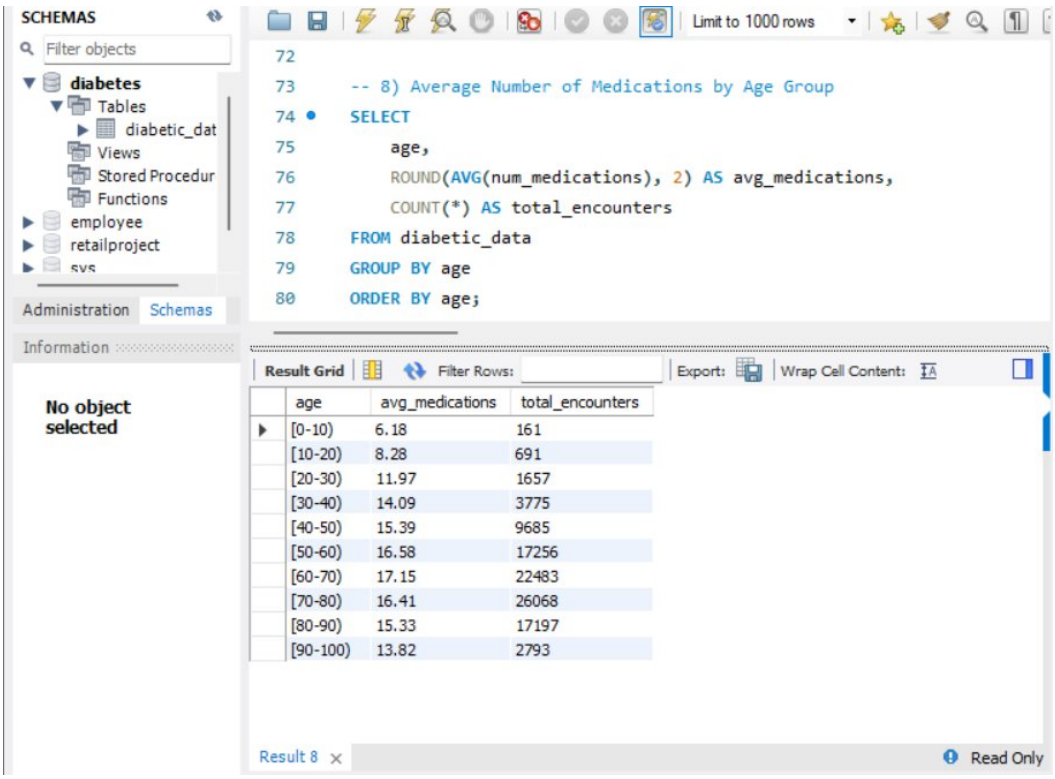
EV-06	SQL – Age Distribution of Patient Encounters



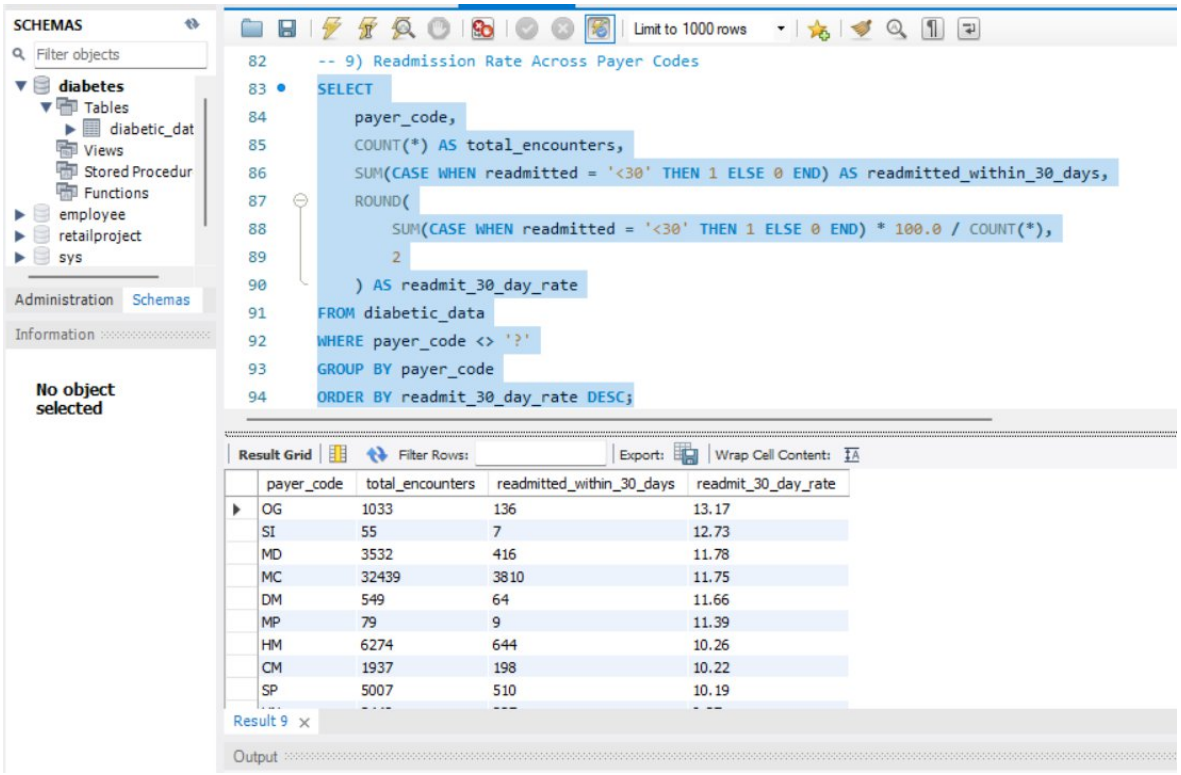
EV-07	SQL – Patient Encounters Procedures



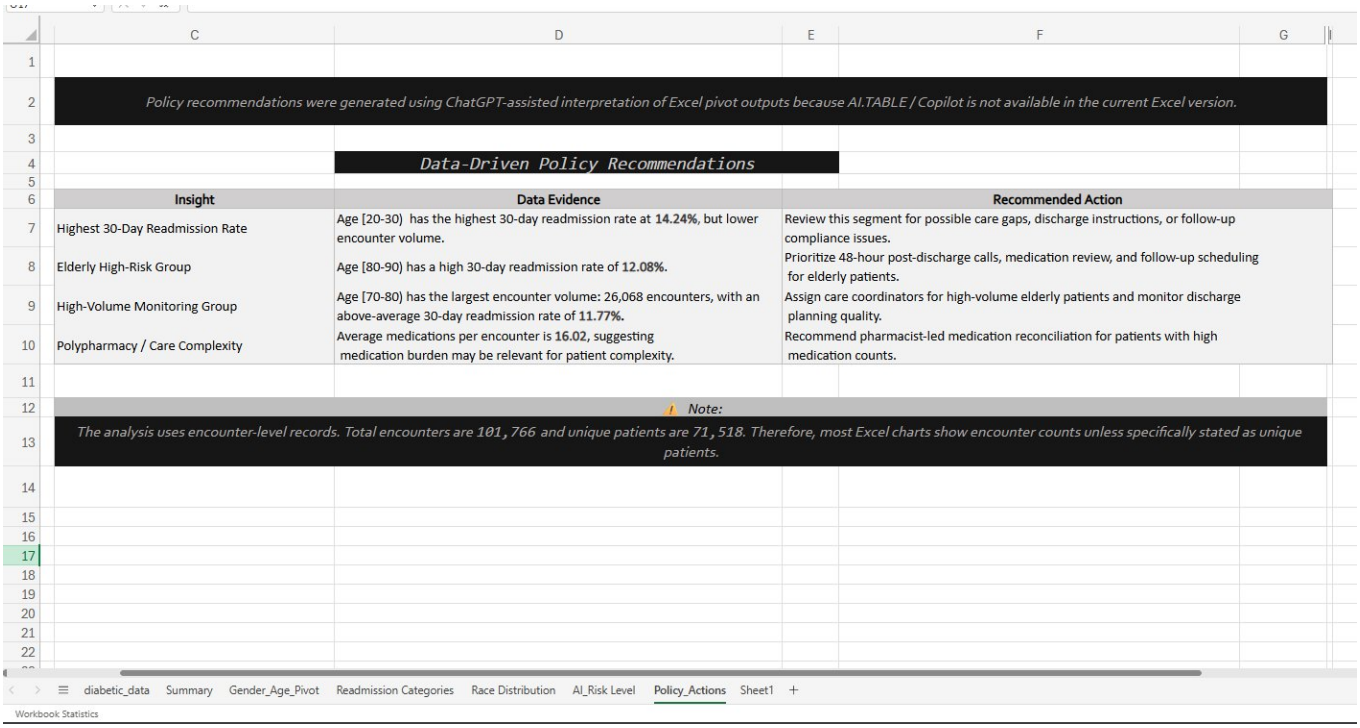
EV-08	Average Medications by Age Group



EV-09	SQL - Readmission Rate by Payer Code



EV-10	Excel – Policy Action Summary



EV-11	Excel – AI-Assisted Policy Recommendations and Risk Summary

⚠ Note:

AI.TABLE requires Microsoft 365 Copilot access, which is not available in the current Excel version. Therefore, an equivalent pivot-based analysis and ChatGPT-assisted summary were used.

PIVOT TABLE ANALYSIS : Age Group Vs 30-Day Readmission Risk

AI-Assisted Age Risk Summary

Count of readmitted	readmitted					
Age group	<30	>30	NO	Grand Total	30-Day Readmission Rate %	Risk Level
[0-10)	3	26	132	161	1.86%	Low
[10-20)	40	224	427	691	5.79%	Low
[20-30)	236	510	911	1,657	14.24%	High
[30-40)	424	1,187	2,164	3,775	11.23%	Medium
[40-50)	1,027	3,278	5,380	9,685	10.60%	Low
[50-60)	1,668	5,917	9,671	17,256	9.67%	Low
[60-70)	2,502	7,897	12,084	22,483	11.13%	Low
[70-80)	3,069	9,475	13,524	26,068	11.77%	Medium
[80-90)	2,078	6,223	8,896	17,197	12.08%	High
[90-100)	310	808	1,675	2,793	11.10%	Low
Grand Total	11,357	35,545	54,864	1,01,766		

The age groups [20-30] and [80-90] show the highest 30-day readmission rates at 14.24% and 12.08%. The [70-80] group also needs attention because it has a large encounter volume and an above-average 30-day readmission rate of 11.77%. These groups should be prioritized for discharge follow-up, medication review, and post-discharge monitoring.

NOTE: *Risk levels are based on the 30-day readmission rate compared with the overall project average of 11.16%. Groups at or above 12% are marked High, groups above the overall average are marked Medium, and the remaining groups are marked Low.*

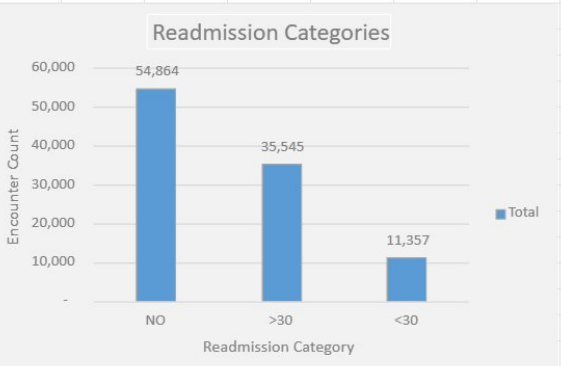
diabetic_data | Summary | Gender_Age_Pivot | Readmission Categories | Race Distribution | AI_Risk Level | Policy_Actions | Sheet1

Workbook Statistics

EV-12	Excel - Detailed Age-Based Risk Summary

Encounter Count by Readmission Category

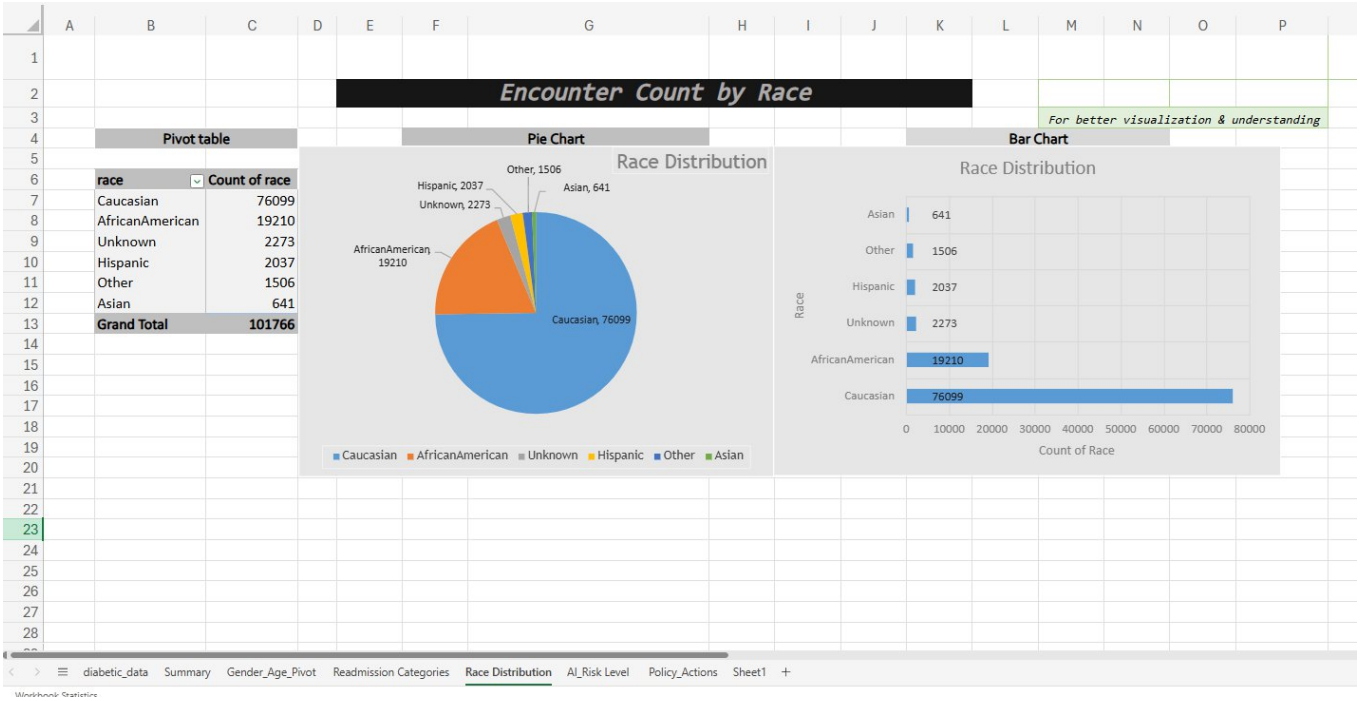
readmitted	Count of readmitted
NO	54,864
>30	35,545
<30	11,357
Grand Total	1,01,766



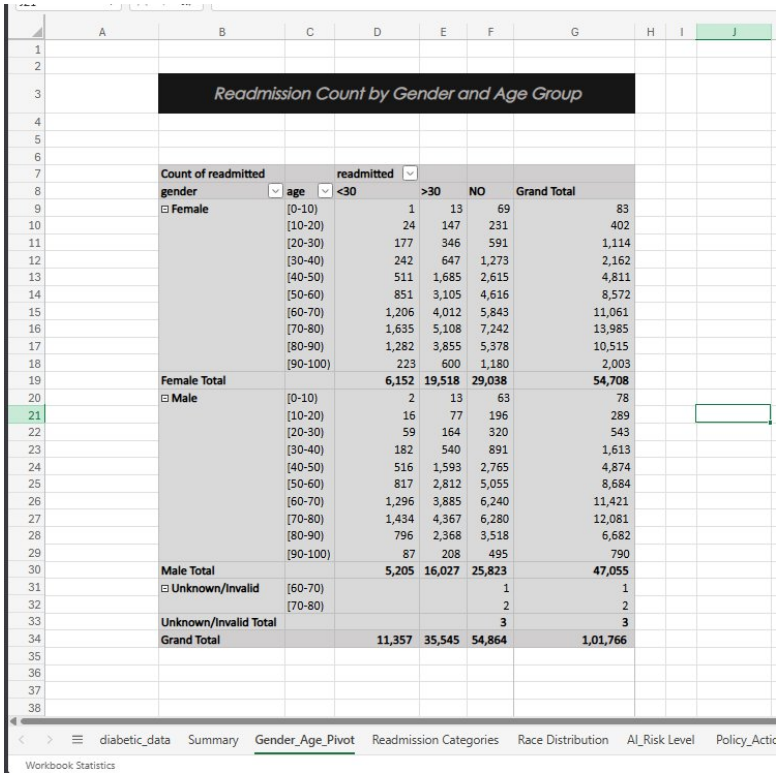
diabetic_data | Summary | Gender_Age_Pivot | Readmission Categories | Race Distribution | AI_Risk Level | Policy_Actions | Sheet1

Workbook Statistics

EV-13	Excel – Readmission Category and Race Distribution



EV-14	Excel - Readmission by Gender and Age Group



Count of readmitted		readmitted			
gender	age	<30	>30	NO	Grand Total
Female	[0-10)	1	13	69	83
	[10-20)	24	147	231	402
	[20-30)	177	346	591	1,114
	[30-40)	242	647	1,273	2,162
	[40-50)	511	1,685	2,615	4,811
	[50-60)	851	3,105	4,616	8,572
	[60-70)	1,206	4,012	5,843	11,061
	[70-80)	1,635	5,108	7,242	13,985
	[80-90)	1,282	3,855	5,378	10,515
	[90-100)	223	600	1,180	2,003
Female Total		**6,152**	**19,518**	**29,038**	**54,708**
Male	[0-10)	2	13	63	78
	[10-20)	16	77	196	289
	[20-30)	59	164	320	543
	[30-40)	182	540	891	1,613
	[40-50)	516	1,593	2,765	4,874
	[50-60)	817	2,812	5,055	8,684
	[60-70)	1,296	3,885	6,240	11,421
	[70-80)	1,434	4,367	6,280	12,081
	[80-90)	796	2,368	3,518	6,682
	[90-100)	87	208	495	790
Male Total		**5,205**	**16,027**	**25,823**	**47,055**
Unknown/Invalid	[60-70)			1	1
	[70-80)			2	2
Unknown/Invalid Total				**3**	**3**
Grand Total		**11,357**	**35,545**	**54,864**	**1,01,766**

EV-15	Excel – Key Project Metrics Summary

Summarized key metrics

Metric	Value
Total Encounters	1,01,766
Unique Patients	71,518
Avg Lab Procedures	43.10
Avg Medications	16.02
Avg Time in Hospitals (days)	4.40
Total Emergency Visits (prior year)	20,133
Total Inpatient Visits (prior year)	64,679
30-Day Readmission Rate	11.16%

Note:

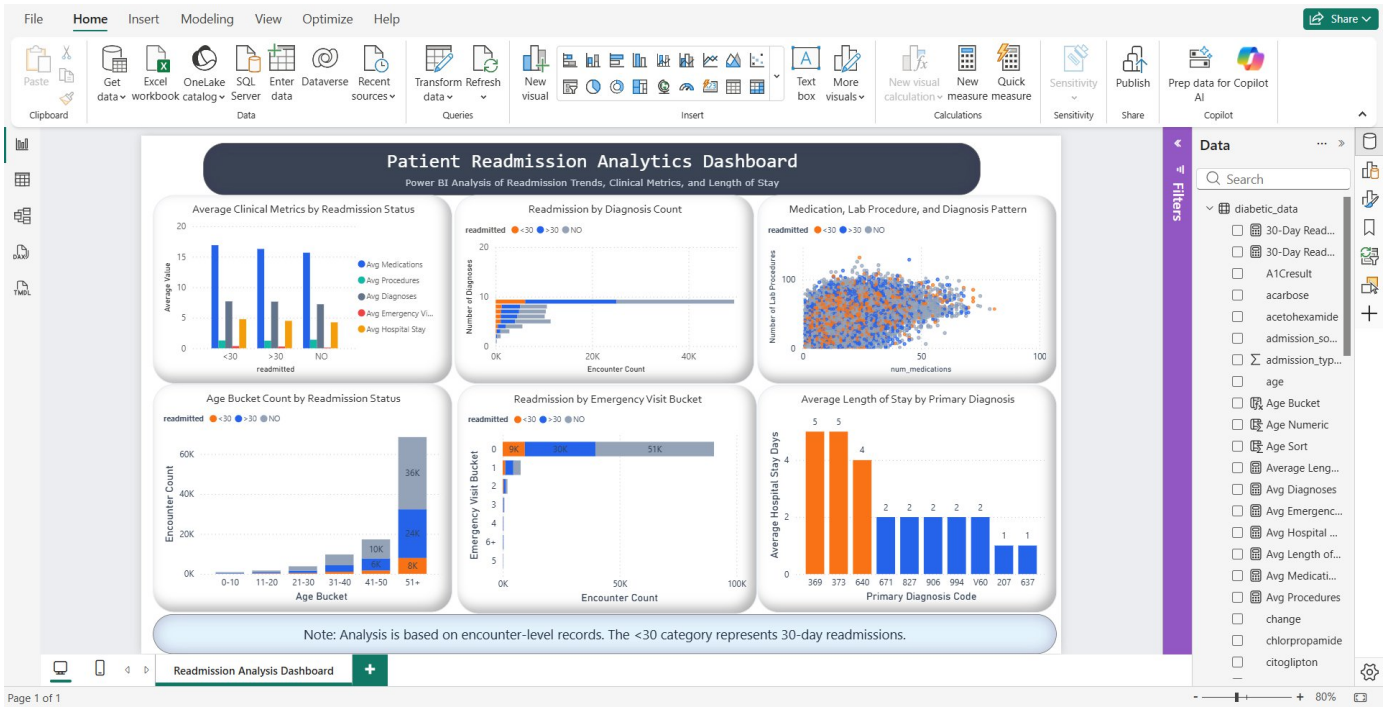
The analysis uses encounter-level records.
Total encounters are 101,766 and unique patients are **71,518.** Therefore, most Excel charts show encounter countsunless specifically stated as unique patients.

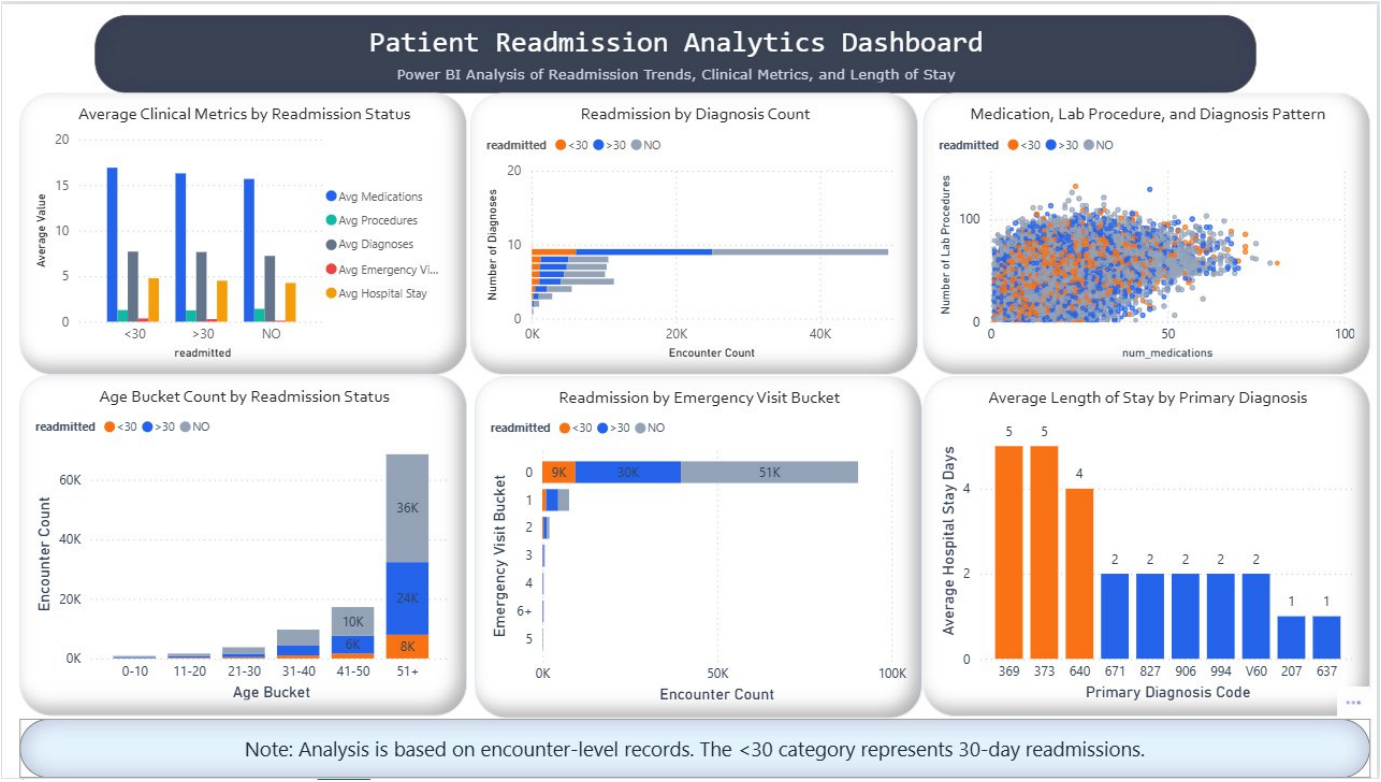
diabetic_data | Summary | Gender_Age_Pivot | Readmission Categories | Race Distribution | AI_Risk Level | Policy_Actions

Workbook Statistics

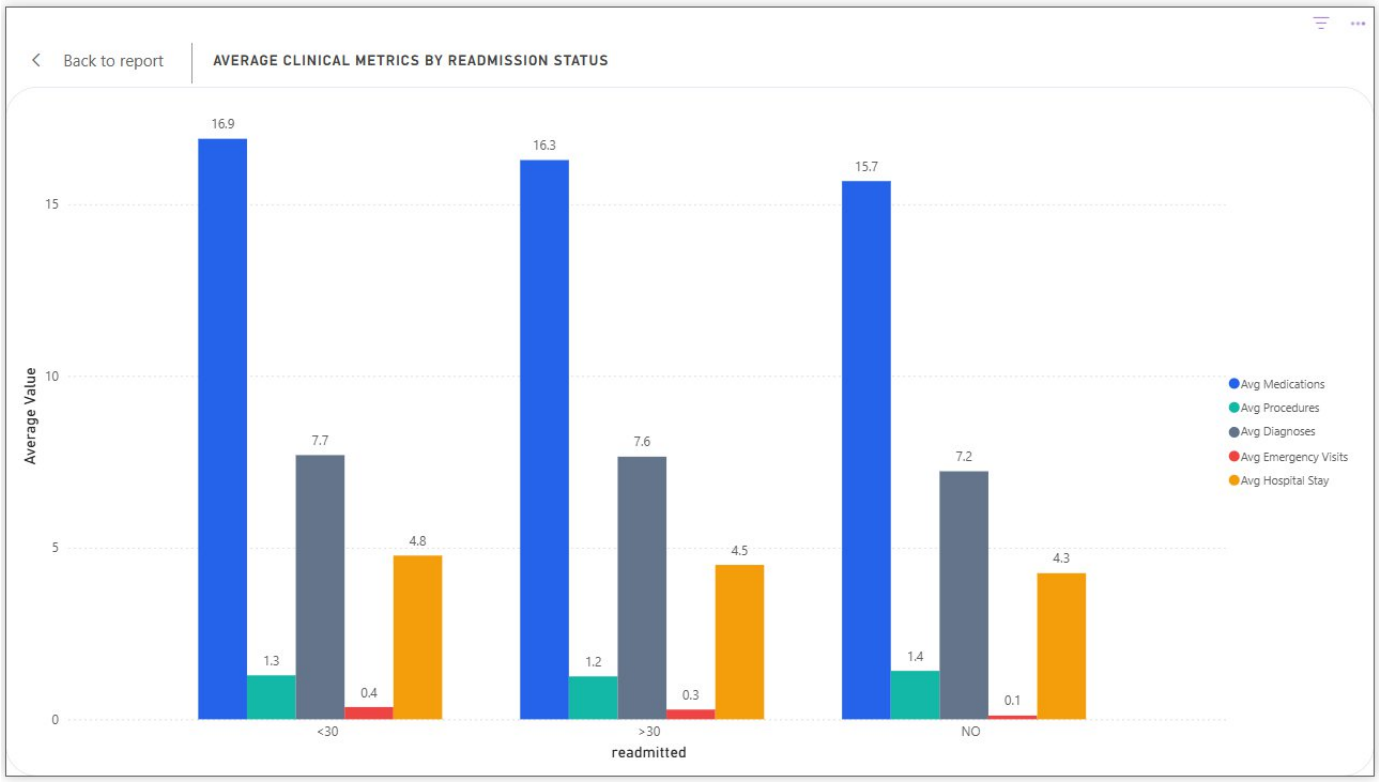
EV-16	Power BI – Dashboard 30-Day Patient Readmission Analysis

The project goal is to identify patients at high risk of being readmitted within 30 days of discharge so healthcare teams can plan early follow-ups, medication reviews, and proactive care interventions.

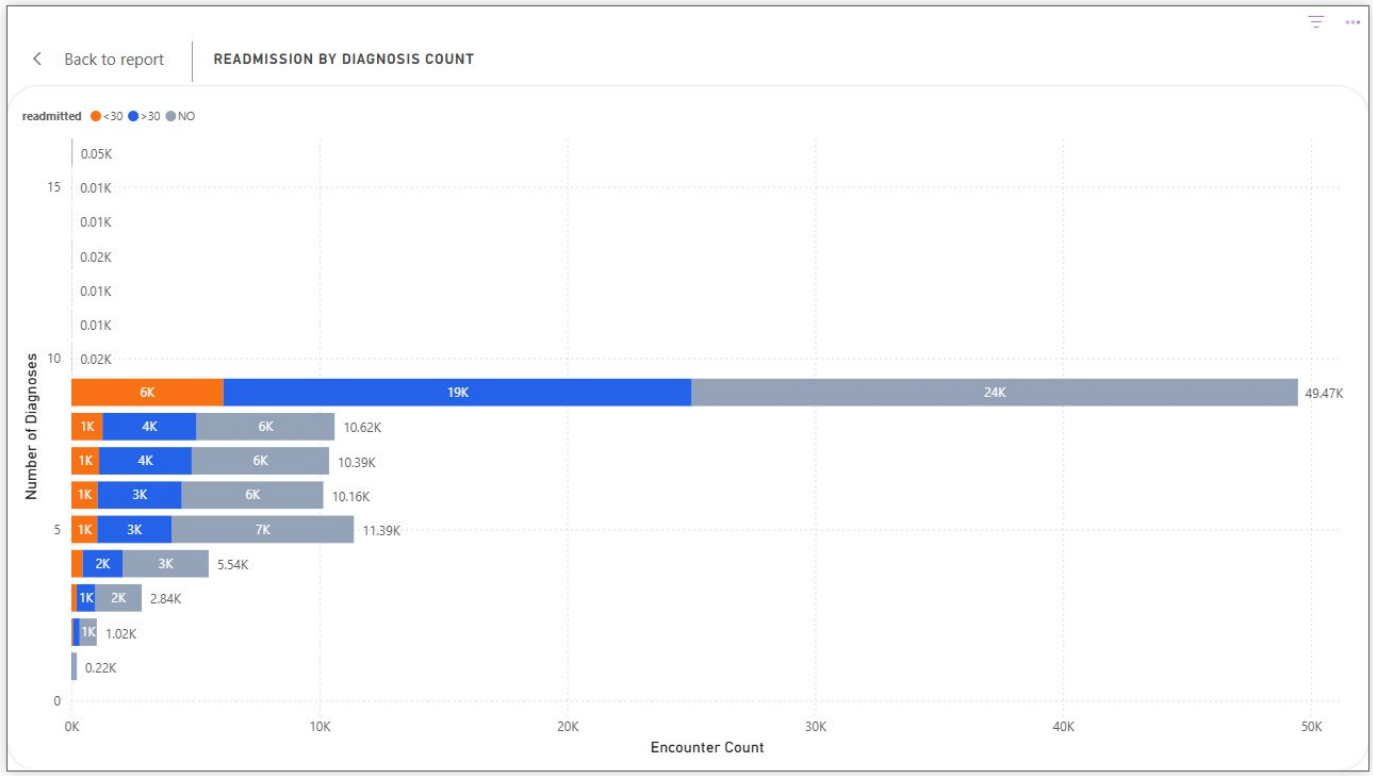




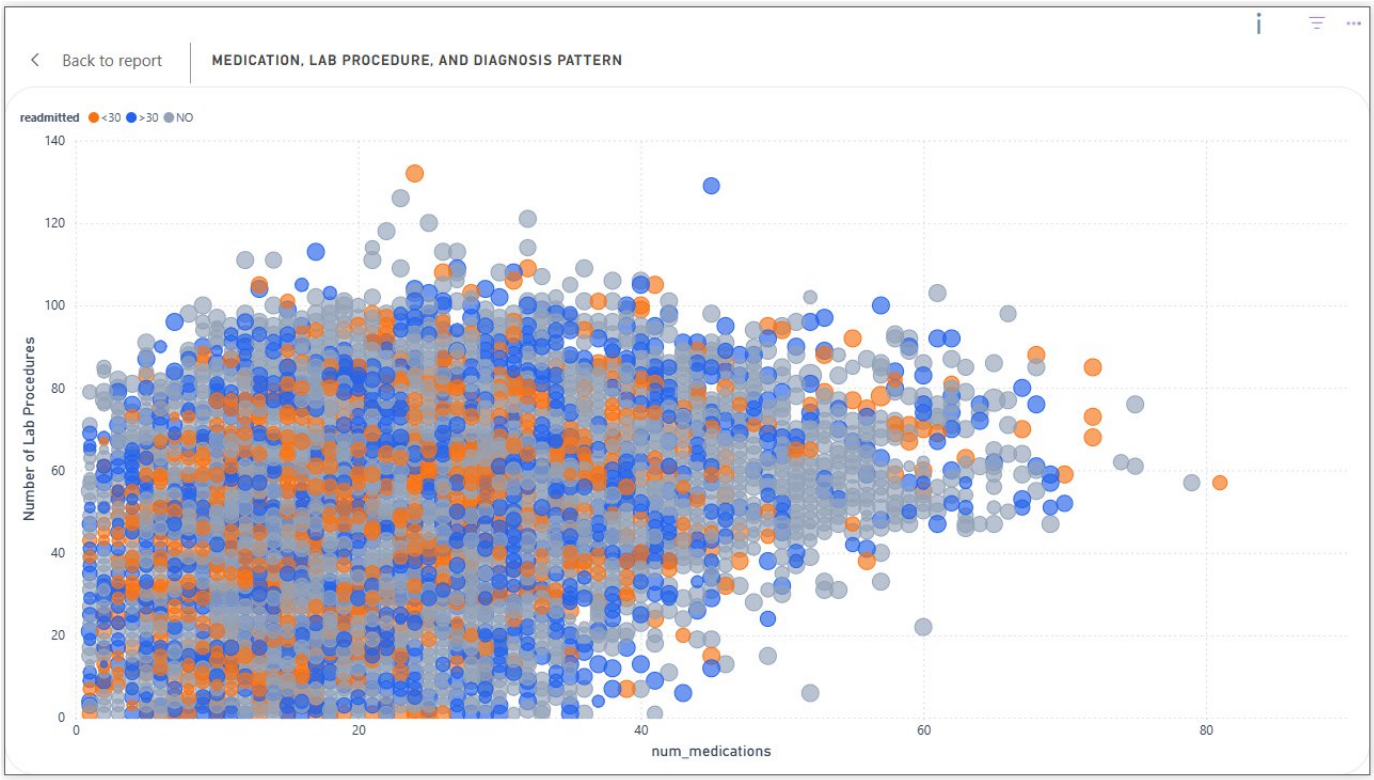
CLINICAL METRICS:



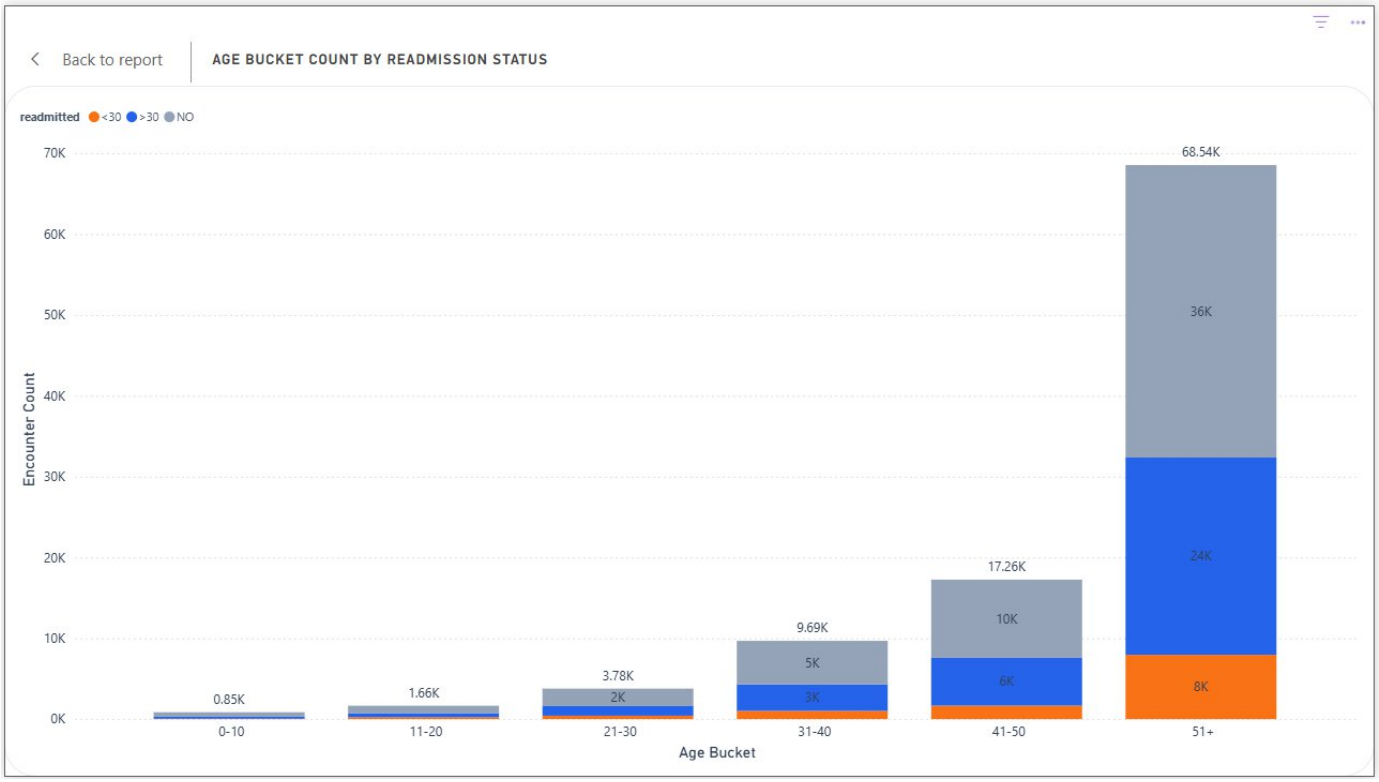
READMISSION BY DAIGNOSIS COUNT:



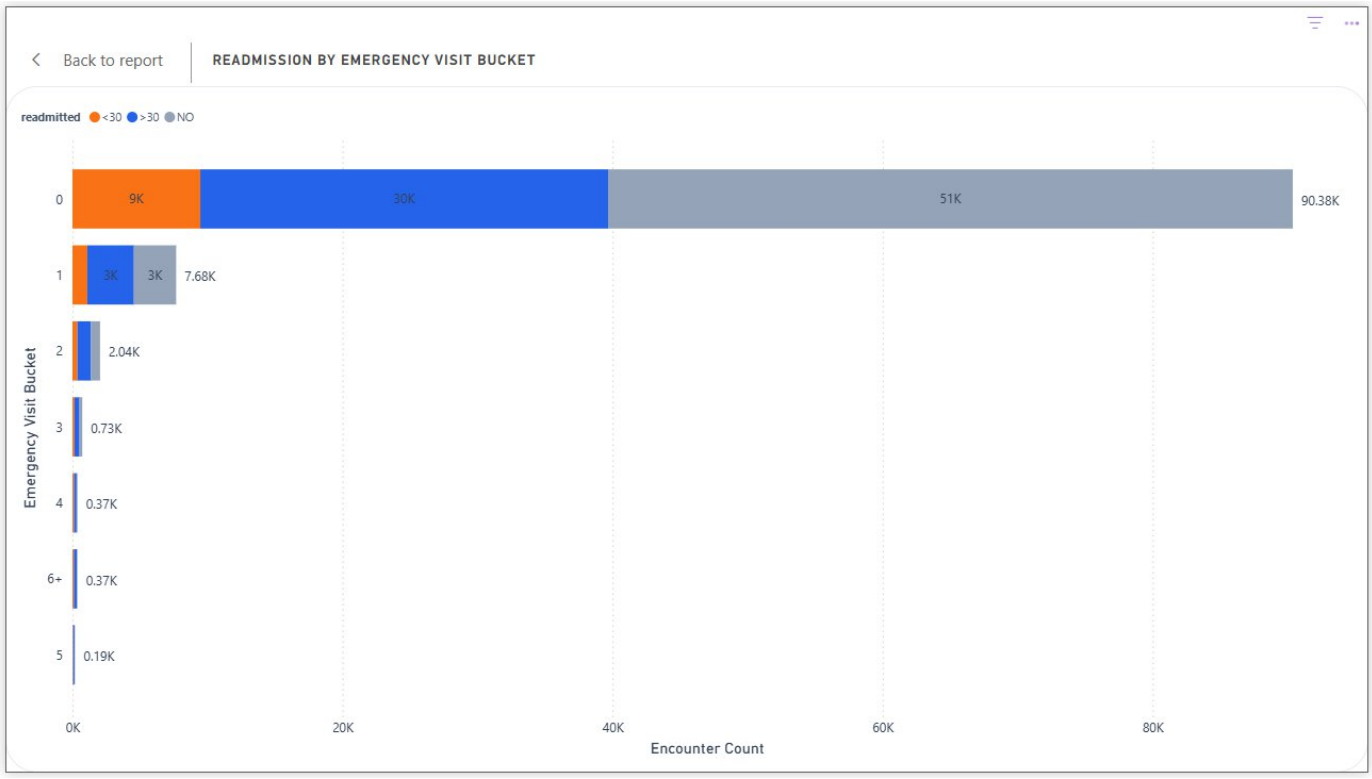
DIAGNOSIS PATTERN:



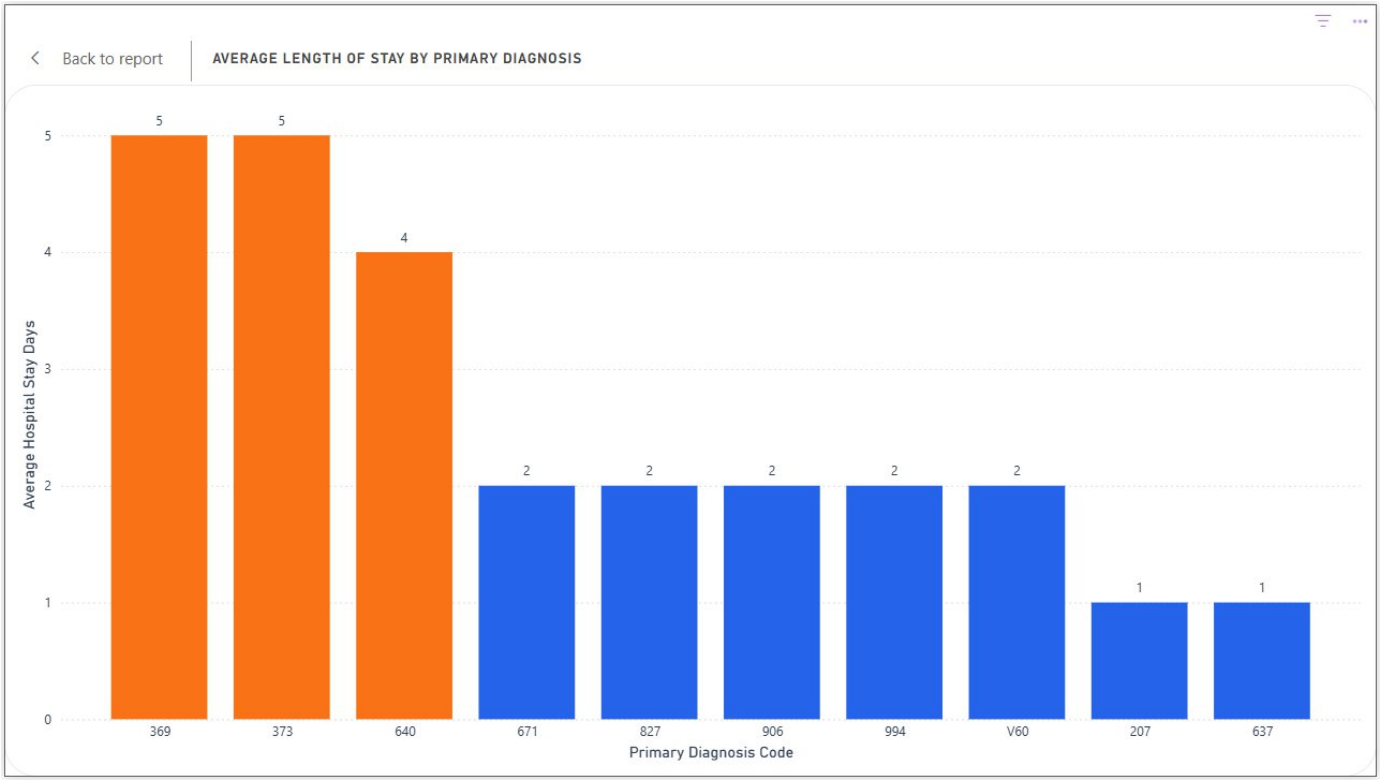
AGE BUCKET:



EMERGENCY VISIT:



AVERAGE LOS IN HOSPITAL:


Back to report
AVERAGE LENGTH OF STAY BY PRIMARY DIAGNOSIS
Average Hospital Stay Days
0
1
2
3
4
5
5
5
4
2
2
2
2
2
1
1
369
373
640
671
827
906
994
V60
207
637
Primary Diagnosis Code

Submission Conclusion

Final evaluator-ready summary

Completed Work	BRD prepared separately, SQL analysis executed, Excel summaries/pivots created, AI-assisted insights documented, and Power BI dashboard prepared.
Primary Insight	Early readmission is concentrated in the <30 category, with 11.16% of encounter records falling into the 30-day readmission group.
Risk Factors Reviewed	Age group, diagnosis count, medication load, procedure count, emergency visits, inpatient history, payer code, and hospital stay duration.
Business Use	Supports proactive care planning, discharge follow-up prioritization, medication review, and policy-level reporting.
Recommended Upload Field	Upload this document under Screenshots/Evidence after converting to PDF, as the portal accepts PDF.

Final Note:

The screenshots and explanations show not only that the outputs were created but also why each output matters for business analysis and healthcare decision-making.